NeuroScan AI — ICAIN 2026 Presentation Script

Total time: 8 minutes · 4 presenters · 2 minutes each · 11 slides

Presenter	Slides Covered	Sections	Time
Student 1	1 – 4	Title, Index, Abstract, Introduction	2:00
Student 2	5 – 6	Literature Review, Research Gaps	2:00
Student 3	7	Proposed Methodology (Architecture Diagram)	2:00
Student 4	8 – 11	Results & Discussion, Comparative Analysis, Conclusion & Future Work, References	2:00

How to use this document: each section below is one presenter's complete speaking script, written to be read almost verbatim. [Slide cues] in italics tell you when to advance or point at something on the slide. Practice each part against a timer once — these scripts are calibrated to land close to 2:00 at a normal, unhurried conference-speaking pace (~130–140 words/minute).

Student 1 — Slides 1–4 (Title, Index, Abstract, Introduction)

[Target time: 2:00. Approx. split — Title 15s, Index 15s, Abstract 45s, Introduction 45s.]

Slide 1 — Title

[Slide 1 on screen]

Good [morning/afternoon], everyone. We are presenting “NeuroScan AI: A Heterogeneous Multi-Task Learning Framework for EEG-Based Neurological Risk Classification,” by Shlok Shah, Rajvi Mahidhar, Dhriti Jangla, Chahak Daga, and Pravin Shrinath, from MPSTME, SVKM's NMIMS.

Slide 2 — Index

[Advance to Slide 2]

Here's how our talk is structured: abstract and introduction, literature review and research gaps, our proposed methodology, results, a comparison with existing work, and conclusion.

Slide 3 — Abstract

[Advance to Slide 3]

EEG is a non-invasive, accessible way to study neurological disorders. But almost all deep learning research trains one model on one dataset for one disorder — and because EEG datasets differ in channel count, sampling rate, and format, these models don't transfer across disorders.

NeuroScan AI unifies risk classification for three structurally incompatible disorders — ADHD, epilepsy, and Alzheimer's — behind one shared representation: dataset-specific preprocessing, adaptive input projection, a shared CNN-BiLSTM encoder, disorder-specific heads, and dataset-balanced batch sampling.

On held-out validation subjects, our corrected model achieved 79.2% accuracy for ADHD, 76.9% for Alzheimer's, and 93% for epilepsy — with Grad-CAM explaining which channels and moments drove each prediction.

Slide 4 — Introduction

[Advance to Slide 4]

The problem: EEG datasets differ in channel count, layout, sampling rate, and labeling — so a model trained on one disorder's dataset cannot transfer to another's.

Our approach is to learn ONE shared EEG representation across three incompatible datasets, instead of three disconnected single-disorder models.

Our contributions: a dataset-specific preprocessing pipeline with adaptive input projection across all three sources; a shared CNN-BiLSTM backbone producing one 256-dimensional embedding for every disorder; a structure-aware masked multi-task objective with dataset-balanced batch sampling; and a Grad-CAM explainability module adapted for EEG, giving channel-and-time attribution per prediction.

[Hand off to Student 2 for Literature Review]

Student 2 — Slides 5–6 (Literature Review, Research Gaps)

[Target time: 2:00. Approx. split — Literature Review 60s, Research Gaps 60s.]

Slide 5 — Literature Review

[Slide 5 on screen — point to the table as you go through rows]

We reviewed recent work from 2024 and 2025 across CNN-based classification, graph neural networks, attention-based Bi-LSTM, Siamese networks, and domain-adversarial seizure detection.

EEGConvNeXt demonstrated CNN-based EEG classification with explainability, but only for Alzheimer's and frontotemporal dementia. A Graph Neural Network approach modeled both Alzheimer's and epilepsy — but as two completely separate models and datasets, not a shared representation. Attention-based Bi-LSTM work optimized specifically for Alzheimer's diagnosis. A Siamese deep neural network handled ADHD detection using a transformed, ADHD-specific representation. And CNN-BiLSTM with domain-adversarial training tackled epilepsy specifically, but doesn't generalize beyond it.

The pattern across all of this literature is consistent: every single reviewed method targets ONE disorder on ONE dataset. No shared multi-disorder representation exists anywhere in the literature we reviewed.

Slide 6 — Research Gaps

[Advance to Slide 6]

This leads us to four specific research gaps. First, most studies focus on single-disorder detection with completely separate models for Alzheimer's, epilepsy, or ADHD.

Second, EEG representations vary widely across the literature — spectrograms, power spectral density maps, wavelet features, graph structures — with no common raw-EEG standard anyone agrees on.

Third, graph-based, domain-adversarial, and explainable methods each solve a narrow, specific problem, but none of them propose a shared framework that works across multiple disorders.

And fourth, review articles in this space explicitly flag an open debate on data variability, generalization, and scalability across disorders — but none of them actually propose a unifying architecture to address it. That's exactly the gap NeuroScan AI is built to fill.

[Hand off to Student 3 for Proposed Methodology]

Student 3 — Slide 7 (Proposed Methodology)

[Target time: 2:00. This is one dense slide — take your time walking the diagram left to right.]

Slide 7 — Proposed Methodology

[Slide 7 on screen — point to each box as you describe it, moving left to right along the diagram]

This diagram is the heart of our architecture. Let me walk through it, left to right.

We start with three heterogeneous EEG sources — ADHD, epilepsy, and Alzheimer's — each with a different channel count and native sampling rate, so none can feed into the same network as-is.

First, preprocessing plus adaptive input projection: every dataset is filtered, resampled to a common 128 Hertz, and normalized. The two 19-channel datasets pass through unchanged, but epilepsy's 23 bipolar channels go through a learned convolutional projection down to 19 — so every dataset now feeds the network in exactly the same shape.

That feeds our shared CNN-BiLSTM backbone, highlighted in blue as our core contribution. A three-block convolutional encoder extracts spatial features within each window, and a bidirectional LSTM learns patterns across windows over time, together producing one single 256-dimensional embedding — the same embedding space for all three disorders.

That embedding feeds three disorder-specific heads. Only the head matching the current batch's dataset gets updated at each step; the other two are masked out entirely — structure-aware masked optimization.

And here's our key engineering fix: dataset-balanced batch sampling. ADHD and Alzheimer's contribute roughly one sample per recording, while epilepsy contributes one per 24-second epoch — thousands more raw rows. Sampled naively, epilepsy drowns out the other two tasks. Our sampler guarantees all three datasets contribute an equal number of batches per epoch — unit-verified at exactly 78 each. This is the exact fix that recovers our ADHD head from complete collapse, which you'll hear more about next.

Finally, we apply Grad-CAM to every prediction — giving channel-by-time attribution, so every risk score comes with an explanation of which electrodes and moments drove it.

[Hand off to Student 4 for Results & Discussion]

Student 4 — Slides 8–11 (Results, Comparative Analysis, Conclusion, References)

[Target time: 2:00. Approx. split — Results 45s, Comparative 35s, Conclusion 30s, References 10s.]

Slide 8 — Results & Discussion

[Slide 8 on screen — point to the table]

Here are our validation results, comparing single-task baselines against our corrected joint model.

For ADHD, the corrected model reached 79.2% accuracy and an F1 of 0.78 — recovering from a complete collapse we saw before the balanced sampler. For Alzheimer's, 76.9% accuracy and an F1 of 0.80, again improving over the pre-correction run. For epilepsy, accuracy stayed high at 93%, but F1 actually dropped from 0.54 to 0.40, driven by a drop in recall.

The honest finding: the same fix that rescues ADHD and Alzheimer's costs epilepsy. An explicit, unresolved trade-off — not something we claim to have fully solved.

Slide 9 — Comparative Analysis

[Advance to Slide 9]

How does this compare to prior work? EEG-ConvNeXt reports up to 99.7% on binary Alzheimer's/FTD classification; ADHD/CD-NET reports around 93.7% internal and 98.2% external accuracy for ADHD. But these are NOT directly comparable to ours — each uses a different dataset, split, and single-disorder task, while we solve a harder, joint three-disorder problem with one shared model. We show these only for context.

Slide 10 — Conclusion & Future Work

[Advance to Slide 10]

To conclude: NeuroScan AI shows one shared CNN-BiLSTM backbone can learn across three incompatible EEG datasets, and that dataset-balanced sampling can recover a collapsed disorder head — while honestly surfacing the trade-off it creates for epilepsy.

We're transparent about our limitations: unverified private ADHD labels, a small 13-subject Alzheimer's validation set, an unisolated sampler/learning-rate confound, and no external validation yet. This is a research prototype, not a deployment-ready tool.

Future work: a controlled ablation isolating the sampler's effect, an epilepsy re-labeling test, gradient accumulation across datasets, and external validation per disorder.

Slide 11 — References

[Advance to Slide 11]

Our full reference list is on screen. Thank you — we're happy to take any questions.